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A Novel High-Performance $TiO_{2-x}/TiO_{1-y}N_y$ Coating Material for Silicon Anode in Lithium-Ion Batteries

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Protective surface coatings on Si anodes are promising for improving the electrochemical performance of lithium-ion batteries (LIBs). Nevertheless, most coating materials have severe issues, including low initial coulombic efficiency, structural fracture, morphology control, and complicated synthetic processing. In this study, a multifunctional $TiO_{2-x}/TiO_{1-y}N_y$ (TTN) formed via a facile and scalable synthetic process is applied as a coating material for Si anodes. A thin layer of amorphous TiO_2 is uniformly coated onto Si nanoparticles by a simple sol–gel method and then converted into a two phase $TiO_{2-x}/TiO_{1-y}N_y$ via nitridation. The lithiated TiO_{2-x} provides high ionic and electrical conductivity, while $TiO_{1-y}N_y$ can improve mechanical strength that alleviates volume change of Si to address capacity fading issue. Owing to these synergetic advantages, $TiO_{2-x}/TiO_{1-y}N_y$-coated Si (Si@TTN) exhibits excellent electrochemical properties, including a high charge capacity of 1650 mA h g^{-1} at 0.1 A g^{-1} and 84% capacity retention after 100 cycles at 1 A g^{-1}. Moreover, a significantly enhanced rate performance can be achieved at a high current density. This investigation presents a facile and effective coating material to use as the high-capacity silicon anode in the emerging Si anode technology in LIBs.

1. Introduction

Silicon (Si) is considered as a promising candidate for anodes of lithium-ion batteries (LIBs) anodes due to its low operation potential (<0.5 V versus Li/Li^+) and high theoretical capacity (3579 mA h g^{-1} for $Li_{3.75}Si$ at room temperature).[1,2] However, Si anodes suffer from large volume expansion, which causes pulverization of the active material, electrical isolation, and the substantial formation of unstable solid–electrolyte interphase (SEI) films during cycling.[3–5] This combination continues to create more SEI on the freshly exposed Si surface, which quickly fills the electrode pores and reduces the battery capacity.[6,7]

To solve the volume expansion issues of Si, many nanostructures and composites have been studied in recent decades. Various material designs such as Si nanowires, nanotubes, nanorods, and porous spheres have been demonstrated to effectively reduce the internal stress of Si anodes during lithiation and show great promise for extending their cycle lifetimes.[8–11] However, these materials are fabricated using toxic reagents or high-cost chemical vapor deposition method. Other designs to accommodate the volume change are core–shell and yolk–shell structures.[12–15] Surface coating with functional layers such as carbon-based materials and metal oxides can improve the electrode conductivity and avoid direct contact between the active material and electrolyte, resulting in the growth of a stable SEI layer.[16] Despite these advantages, these structures still cannot endure the volume changes of Si or have poor initial/late cycle coulombic efficiencies (CEs), which hinder the development of various Si-based composites for practical applications.

Herein, we report a novel multifunctional hybrid coating material for Si anode, $TiO_{2-x}/TiO_{1-y}N_y$ (denoted as TTN), which shows electrically and mechanically robust features. The hybrid coating layer offers multiple advantages. i) The TiO_{2-x} phase provides electrical pathways for electrochemical reactions between Li ions and Si. While white TiO_2 shows an initially low electrical conductivity, black (oxygen-deficient) TiO_2 has a relatively high electrical conductivity due to its narrower bandgap.[17–19] During lithiation, TiO_{2-x} forms the Li_zTiO_{2-x} phase, which acts as an electron and Li ion pathway.[20,21] ii) $TiO_{1-y}N_y$, which is regarded as a solid solution of TiN and TiO, can effectively withstand the volume expansion of Si because of its much larger bulk modulus than TiO_2.[22–25] Moreover, nitrogen atoms incorporated into the oxygen vacancies of the TiO phase improve the electrical conductivity of Si.[26] iii) The hybrid coating layer can also enhance the structural stability of the Si composite because the hybrid coating layer acts as an interfacial barrier between the Si NPs and electrolyte, reducing the creation of an unstable SEI layer. Based on these properties, $TiO_{2-x}/TiO_{1-y}N_y$-coated Si (Si@TTN) may be promising in high-performance Si-based LIBs.

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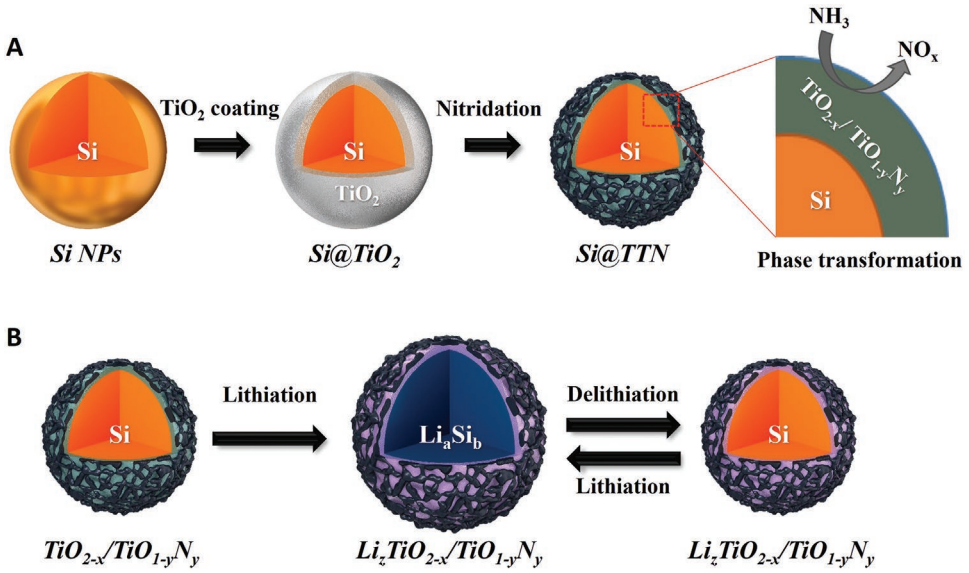

Figure 1. Schematic illustrations of the A) synthesis of Si@TTN nanocomposite and B) phase transformation of the coating layer through electrochemical reaction.

2. Results and Discussion

The process flow for synthesizing the composite is shown in **Figure 1**A. To obtain Si NPs coated with an amorphous TiO_2 thin film, the sol–gel method was adopted with a precursor solution prepared using titanium isopropoxide and Si NPs. The coating reaction was conducted in a mixed solvent of ethanol and aqueous ammonia, which contributes the uniform growth of amorphous TiO_2.[27] After a sufficient reaction, Si@TiO_2 nanocomposite can be obtained. Si@TiO_2 was thermally annealed in a tube furnace at 650 °C for 3 h under flowing NH_3 gas, and the TiO_2 converted into TTN because of the strong reduction ability of NH_3, thereby forming Si@TTN core–shell nanoparticles. Figure 1B shows the phase transformation of TTN during the lithiation/delithiation. During the lithiation, TiO_{2-x} formed Li_zTiO_{2-x} phase while $TiO_{1-y}N_y$ component remained in its original phase. This is because lithium ions can be intercalated into TiO_{2-x}, whereas $TiO_{1-y}N_y$ is inactive with lithium ions.

Li_zTiO_{2-x} remains even after complete delithiation, which can provide electrical pathways and promote the electron/Li ion conductivity. Additionally, the $TiO_{1-y}N_y$ structure has a high mechanical strength and electrical conductivity, which enhance the structural stability and rate performance of the electrode.

To investigate the phase transformation conditions of TiO_2 according to the nitridation reaction temperature, the commercial TiO_2 powder was used. As shown in Figure S1A (Supporting Information), the color of TiO_2 changed from white to dark blue and finally black at high temperature. This is related to the electronic structure and has been discussed in many reports.[28–30] In the X-ray diffraction (XRD) patterns (Figure S2, Supporting Information), the ratio of $TiO_{1-y}N_y$ phase was increased at the high temperature, which means that $TiO_{1-y}N_y$ contributed to the color change of the powder toward black. The TEM images show that the $TiO_{1-y}N_y$ had a hollow structure, whereas TiO_2 had a filled structure (Figure S1B, Supporting Information). The driving force creating the hollow $TiO_{1-y}N_y$ can be explained from several perspectives. At a low temperature, TiO_2 is reduced by NH_3 gas to yield TiO_{2-x} with an oxygen-deficient structure before the TiO phase can form at a higher temperature. The TiO phase has a large nonstoichiometry, stabilizing its structure by containing randomly distributed vacancies ≈15% on both Ti and O sites.[31] This suggests that oxygen vacancies serve as appropriate sites for N atoms because they are thermodynamically stable for N atom incorporation. The Ti^{2+} (for TiO) species can form stable N^{3-} by donating electrons to N atoms, thereby promoting the substitution of O and N.[32] The O atoms diffuse outward for the anion exchange and the inserted N atoms reduce the energy needed to form oxygen vacancies that diffuse inward. Because of imbalance of the diffusion rate, anion exchange occurs near the particle surface and oxygen vacancies merge to form voids. Additionally, density-functional theory calculations have indicated that the incorporated N atoms must be stabilized from N^{2-} to N^{3-}, allowing for the Ti 3d electrons in the TiO conduction band to transfer to the N atoms.[24] In the XRD patterns of nitrided TiO_2, shift of the peak toward lower angle was observed indicating that TiN-rich phase is formed at high temperature (Figure S2A, Supporting Information). As illustrated by the plotted composition ratios (Figure S2B, Supporting Information), both TiO_{2-x} and $TiO_{1-y}N_y$ phases coexisted below 800 °C, while only the $TiO_{1-y}N_y$ phase was present above 800 °C. Based on the above analysis, it was confirmed that TiO_2 nitrided at 650 °C had hybrid structure containing TiO_{2-x} and $TiO_{1-y}N_y$ in a weight ratio of ≈7:3.

The Si NPs exhibit smooth surfaces with dimensions of tens of nanometers (Figure S3, Supporting Information). After TiO_2 coating, Si@TiO_2 particles show slightly rough surfaces but maintain their spherical morphology containing crystalline Si cores, as determined by TEM analysis (**Figure 2**A). The fast Fourier transform (FFT) pattern obtained from the TEM image indicates the amorphous nature of the TiO_2 coating layer, which had a thickness of ≈3.5 nm. When Si@TiO_2 was annealed at

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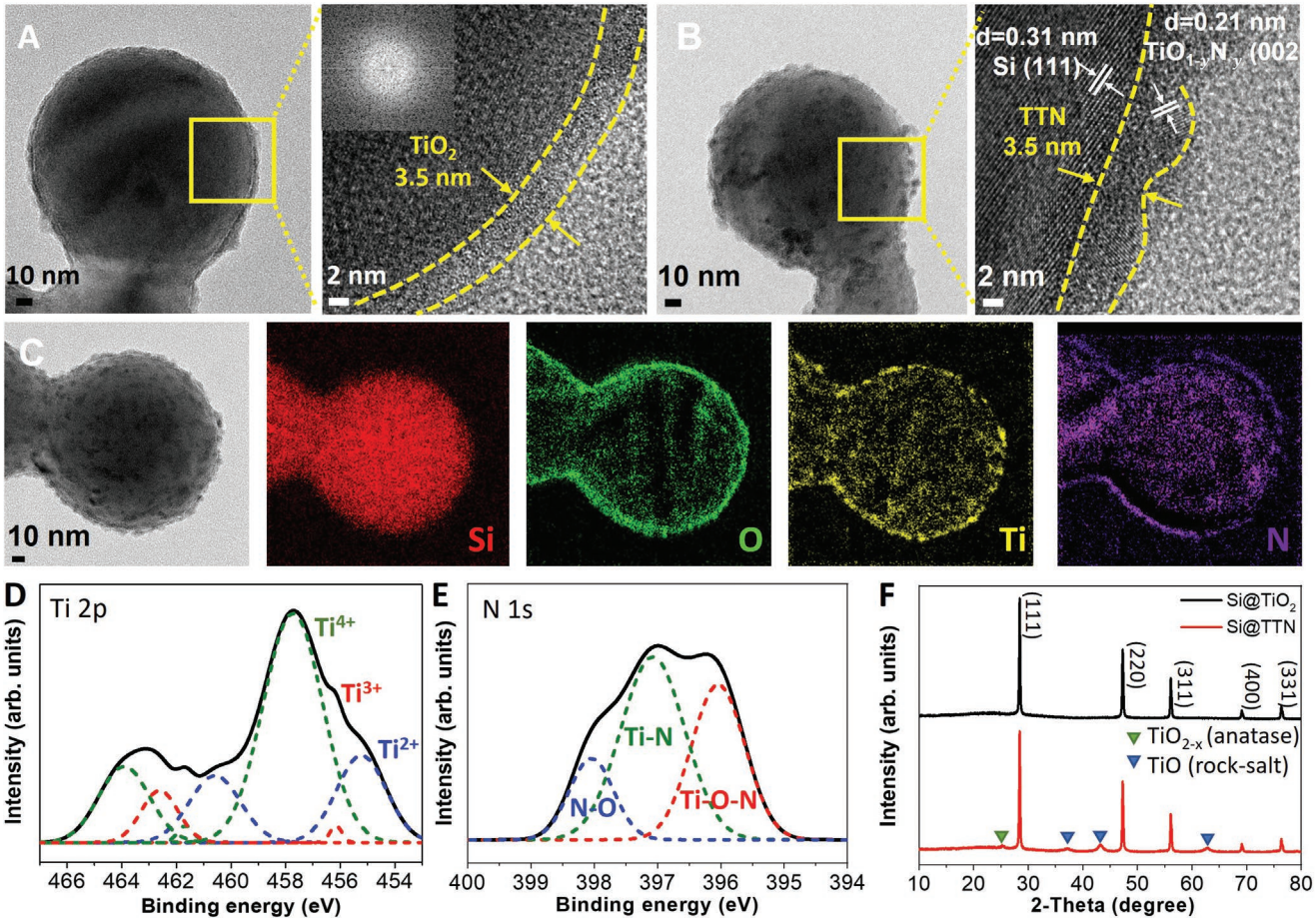

Figure 2. HR-TEM images of A) Si@TiO_2 and B) Si@TTN at low and high magnifications showing the coating layer thickness of TiO_2 and TTN, with the FFT in the inset. C) EELS elemental maps of Si@TTN with respect to Si, O, Ti, and N. D) Ti 2p and E) N 1s XPS spectra of Si@TTN. The dotted lines indicate the fitted data. F) XRD patterns indicating the formation of the TiO phase after nitridation.

650 °C in an NH_3 gas atmosphere, TiO_2 was converted to TTN phase (Figure 2B). The further roughened surface than TiO_2 coating layer was derived from crystalline growth of $TiO_{1-y}N_y$, which was impregnated by TiO_{2-x} film, and higher resolution TEM image (Figure 2B inset) shows lattice fringes in the shell layer with a *d*-spacing of 0.21 nm, corresponding to the $TiO_{1-y}N_y$ (002) orientation. In the electron energy loss spectroscopy (EELS) elemental mapping images of Si@TTN in Figure 2C, red, green, yellow, and purple colors indicate the existence of Si, O, Ti, and N atoms, respectively. The O, Ti, and N signals overlap indicating that Si was embedded uniformly in the TTN coating layer. The photograph of Si@TiO_2 and Si@TTN powders in Figure S4 (Supporting Information) show the color change after nitridation demonstrating the phase transformation to TTN. The XPS spectra are provided for the Ti 2p and N 1s regions in Figure 2D,E. The spectrum of Si@TTN contained a wide range of peaks in the Ti 2p region, while Si@TiO_2 showed only two sharp peaks (Ti $2p_{3/2}$ and Ti $2p_{1/2}$) indicating the Ti^{4+} oxidation state (Figure S5A, Supporting Information). The oxidation state of Ti^{3+} is attributed to the TiN phase, and that of the Ti^{2+} species indicates the TiO phase. We also note that Ti 2p spectrum of Si@TiO_2 revealed peaks at 464.8 and 459.3 eV corresponding to Ti^{4+}; however, those of Si@TTN were observed at 463.9 and 457.7 eV. The peak shifts toward lower binding energy indicating a lower oxidation state of Ti, which is ascribed to oxygen vacancies for TiO_{2-x} in TTN. The existence of nitrogen in Si@TTN composite was evidenced by the N 1 s region with peaks corresponding to Ti-N (397.07 eV) and Ti-O-N bonds (395.4 eV), while no noticeable peaks corresponding to the nitrogen bonds were observed in Si@TiO_2 (Figure S5B, Supporting Information). These results can demonstrate the successful formation of the hybrid coating layer.

The amorphous property of TiO_2 was established by the XRD patterns (Figure 2F), in which only the diffraction peaks from crystalline Si were observed with no signals corresponding to TiO_2. In the XRD pattern of Si@TTN composite, new peaks were observed at 25° corresponding to the (101) plane of anatase TiO_2 and 37°, 43°, and 63° corresponding to the (111), (200), and (220) planes of rock-salt TiO, respectively. The diffraction peaks from Si NPs also observed at 28.4°, 47.3°, 56.1°, 69.1°, and 76.3° corresponding to the (111), (220), (311), (400), and (331) planes, respectively.

In the electrochemical testing of the synthesized composites (**Figure 3**A), Si@TTN exhibited a reversible capacity of 1650 mA h g^{-1} at 0.1 A g^{-1} and an initial Coulombic efficiency of 76%, which exceeds that of Si@TiO_2 (72%) because TiO_2

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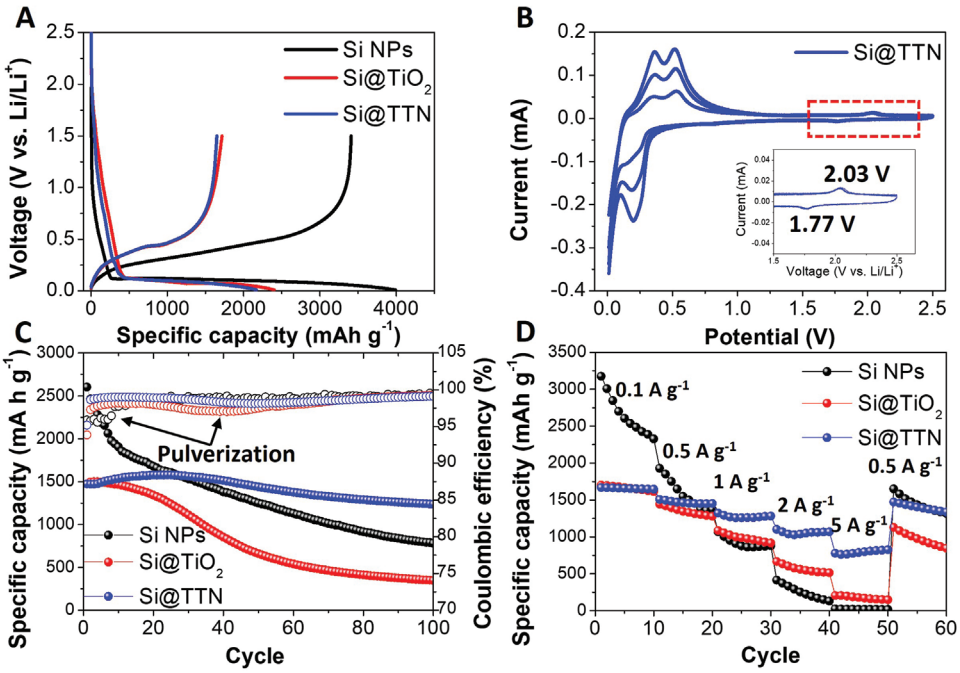

Figure 3. A) Galvanostatic charge/discharge voltage profiles of Si NPs, Si@TiO_2, and Si@TTN measured at 0.1 A g^{-1}. B) CV curves of Si@TTN at a scanning rate of 0.1 mV s^{-1} in the voltage range from 0.01 to 1.5 V. The inset shows an enlargement of the voltage range from 1.5 to 2.5 V. C) Cycling performance and CEs of the Si NPs, Si@TiO_2, and Si@TTN at 1 A g^{-1} for 100 cycles. D) Rate capabilities of the three samples under current densities from 0.1 to 5 A g^{-1}.

irreversibly trapped Li ions whereas $TiO_{1-y}N_y$ in TTN was not reactive. To compare the theoretical reversible capacity of Si@TTN with the above practical value, the core-shell volumetric ratio was calculated. The core and shell thickness were used from the TEM images in Figure 2A,B. The core volume fraction was obtained as the core volume divided by the total core-shell volume, and the calculated value was 67% indicating that the volumetric ratio of Si to TTN was 67:23 (v/v). In order to convert the above result into a weight fraction, the density of each component in Table S1 (Supporting Information) was used and it was obtained as 49:51 (wt/wt). Thus, the calculated theoretical capacity was 1670 mA h g^{-1}, which is almost identical to the measured practical capacity of 1650 mA h g^{-1}. These results indicate that the Si core can be completely activated owing to the short diffusion pathway of Li ions and the high electric conductivity of Si@TTN. The slight decrease in practical capacity compared to the calculated value might be an irreversible reaction of TiO_{2-x} with Li ions.

At the cyclic voltammetry (CV) scan of Si@TTN composite, peaks at 1.77 V during discharging and 2.03 V during charging were observed (Figure 3B). These suggest that Li ions can insert into TiO_{2-x}, leading to the formation of Li_zTiO_{2-x}.[20] The Li_zTiO_{2-x} phase can promote Li ion/electron pathways due to the existence of Ti^{3+}, thereby improving the electrochemical performance of Si@TTN anode. To further investigate the phase transformation of TTN during the first cycling, ex situ XRD analysis was conducted as shown in Figure S6 (Supporting Information). When TTN was lithiated by 0.01 V, two peaks appeared at 33.9° and 44.4° compared to the fresh electrode, indicating phase transition of TiO_{2-x} to orthorhombic $Li_{0.5}TiO_{2-x}$.[33,34] Subsequently, delithiated electrode by 1.5 V represented that the $Li_{0.5}TiO_{2-x}$ phase is still remained due to residual Li in TiO_{2-x} during first cycling that can improve ionic conductivity on the following rate or cycle properties.[19,35] On the other hand, the Si nanocomposites with TiO_2 or $TiO_{1-y}N_y$ did not exhibit any peaks at a high voltage (Figure S7, Supporting Information). The absence of oxidation/reduction peaks at a high voltage region in CV curves of TiO_2 results from disordered structure and defects of amorphous TiO_2 eliminating the plateaus of reaction with Li and TiO_2 and that of $TiO_{1-y}N_y$ means its inactive property with Li ions.[36] For the measurement of cycle life, all samples were precycled at 0.1 A g^{-1} in the voltage window of 0.01–1.5 V. Although the Si NPs initially had the highest capacity of 2603 mA h g^{-1}, the capacity decreased drastically to 781 mA h g^{-1} after 100 cycles, corresponding to a cycle retention of only 30% (Figure 3C). Si@TiO_2 exhibited a relatively stable cycling performance up to 20 cycles, but the capacity then sharply dropped, corresponding to a retention of only 24% (348 mA h g^{-1}). The capacity drop can be explained by cracking and disintegration of the TiO_2 coating layer during repeated charging and discharging cycles. In the case of Si@TTN, the charge capacity was 1473 mA h g^{-1} and remained at 84% after 100 cycles. Although the excellent capacity retention of Si@TTN could be attributed partially to the moderate mass loading of the electrode (0.3 mg cm^{-2}), the superior performance of Si@TTN compared to that of Si NPs and Si@TiO_2 under same condition verifies the critical role of the novel design for hybrid coating layers in the cycling performance. Even when the electrode loading increased by 3.3 times (0.3 → 1 mg cm^{-2}), Si@TTN retained 75% of the original capacity after 50 cycles at 1 A g^{-1} (Figure S11, Supporting Information), reconfirming the superior structural advantage of TTN.

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The CE of the Si NPs increased to 95.8% after precycle but maintained low values for the first few cycles. Si@TiO_2 showed a relatively good CE of 97.3% at first and high average CEs of 98.0% between 1 and 20 cycles. However, the CE decreased shortly and recovered slowly because of the aforementioned pulverization of Si. In comparison, Si@TTN exhibited an enhanced CE of 98.7% at first and showed high values during cycling. These results can be explained by the bulk modulus of TiO_2, TiO, and TiN. TiO_2 shows lower bulk modulus of 179 GPa than TiO and TiN (197 and 280 GPa, respectively).[24] Due to the above properties, TiO_2 has relatively low structural stability against volume expansion of Si, resulting in cracking and disintegration of coating layer during cycling, which deteriorate the CEs of anode. Consequently, high cycling retention of Si@TTN is attributable to the rigid structure of $TiO_{1-y}N_y$, which can withstand the stress exerted by Si volume expansion.

The superiority of Si@TTN was also evidenced by the rate capability (Figure 3D). Reversible capacities of 1658, 1646, 1451, 1282, 1070, and 824 mA h g^{-1} were achieved at current rates from 0.1 to 5 A g^{-1}, whereas Si@TiO_2 showed much lower capacities of 1702, 1613, 1280, 917, 513, and 148 mA h g^{-1}. This difference mainly arises from oxygen-deficient TiO_{2-x}, which has a favorable electrical conductivity due to its narrower bandgap than TiO_2 and enhances ionic conductivity after lithiation. Furthermore, the presence of N atoms in the $TiO_{1-y}N_y$ phase affords conductive components to improve the rate performance of the anode. To further understand the effect of the Li_zTiO_{2-x} phase on the electrochemical performance, Si@$TiO_{1-y}N_y$ composite was prepared by completely converting TiO_2 to $TiO_{1-y}N_y$ at a high temperature (Figure S8, Supporting Information). Si@$TiO_{1-y}N_y$ composite shows a similar cycling performance trend to Si@TTN. However, the lower reversible capacities of 1787, 1752, 1518, 1308, 972, and 553 mA h g^{-1} were delivered at current densities of 0.1 to 5 A g^{-1}. This result means that both the Li_zTiO_{2-x} and $TiO_{1-y}N_y$ phases have a positive effect on the rate performance, and the synergistic effect of the hybrid materials can improve the electrochemical performance of Si anodes. Impedance spectroscopy (EIS) shows the excellent electrochemical performances of Si@TTN (Figure S9, Supporting Information). Si@TTN shows a smaller semicircle than Si@TiO_2 in the high frequency region after the first cycle, which corresponds to a low charge transfer resistance (R_{ct}) of Si@TTN. The lower R_{ct} values of Si@TTN are attributed to the high ionic and electrical conductivity of the hybrid coating layer. After 100 cycles, decreased diameter of semicircle of Si@TTN demonstrates its high structural stability even after repeated lithiation/delithiaion while Si@TiO_2 exhibits two semicircles in the high frequency region indicating the formation of thick SEI layer due to the structural failure.[37,38]

To further demonstrate the structural effect of coating, the surfaces of the three different electrodes were compared before and after cycling using field-emission scanning electron microscopy (FE-SEM) (**Figure 4**A,B). All fresh electrodes exhibit spherical nanoparticles as shown in Figure 4A. However, FE-SEM images in Figure 4B indicate that Si NPs and Si@TiO_2 show severe fracture and agglomeration after cycling. The pulverization of Si NPs was inevitable due to the absence of buffer layer at the beginning of the cycle results in the formation of thick SEI layer. The Si@TiO_2 shows the same morphology as

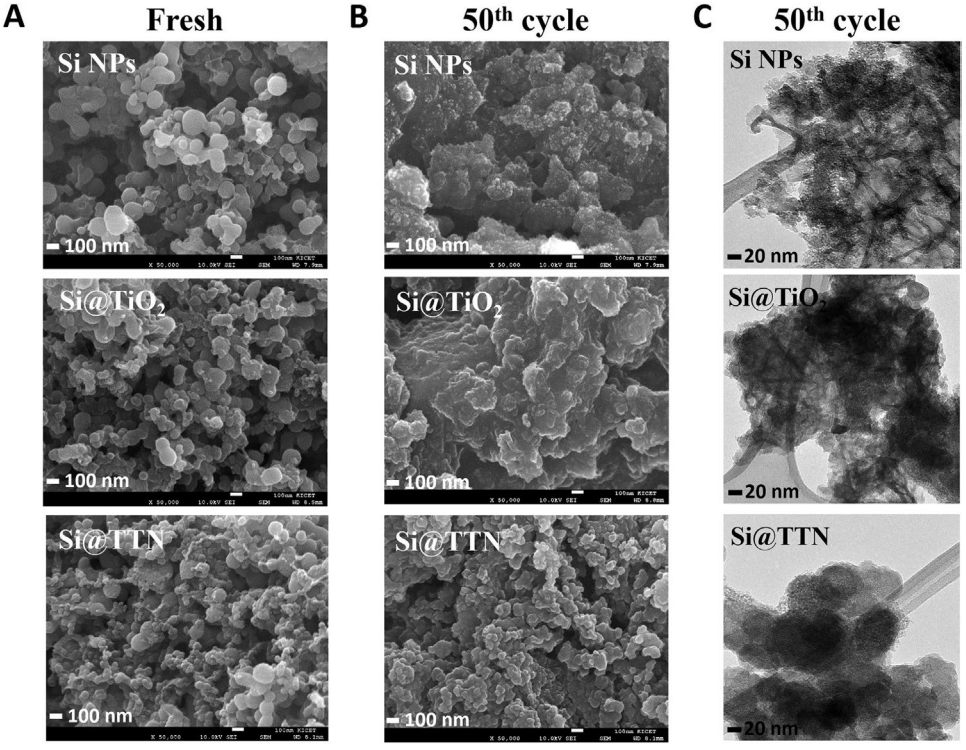

Figure 4. FE-SEM structural assessment of the three different electrodes A) before and B) after cycling at 1 A g^{-1} and C) HR-TEM images showing the structural maintenance of bare Si NPs, Si@TiO_2, and Si@TTN electrode after 50 cycles.

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Si NPs because TiO_2 was collapsed after 20 cycles as previously described in Figure 3C. On the other hand, the spherical morphology of Si@TTN particles was found to be very robust maintaining core-shell structure after cycling (Si@TTN in Figure 4C; Figure S10, Supporting Information). The robust TTN structure of Si@TTN allows to endure the volume expansion of the inner Si NPs while keeping the integration of both components.

3. Conclusion

In conclusion, we have introduced a hybrid TTN coating for Si anodes via a simple coating method to achieve Si-based LIBs with high energy density and long cycle life. The nanostructure was confirmed by various analytical methods, which show that the core-shell composite consisted of Si NPs as the core and a rigid TTN thin film as the shell. Si@TTN composite can achieve a high reversible capacity, stable cycle life, and good rate capability. The improvements are attributed to the synergistic effect of the hybrid coating layer providing enhanced mechanical and electrical properties. In terms of efforts for the commercialization of Si-based anodes, Si@TTN composite represents a promising candidate for high-performance LIBs.

4. Experimental Section

Sample Preparation: Silicon nanoparticles (Si NPs, 99%, average diameter = 50 nm, NanoAmor) and titanium isopropoxide (TTIP, 97%, SIGMA ALDRICH) were used as starting materials. The amorphous TiO_2 was deposited on the surface of the Si NPs via a sol-gel reaction. Typically, 0.5 g of Si NPs was added to a solution containing 666 mL of absolute ethanol and 8 mL of an ammonia aqueous solution (5 N). Afterward, 1.9 mL of TTIP was added dropwise, and the mixed solution was stirred at 40 °C for 40 h. The reaction products were centrifuged at 11 000 rpm for 30 min and washed with ethanol before drying at 80 °C overnight. The amorphous TiO_2 thin film was then converted into $TiO_{2-x}/TiO_{1-y}N_y$ by nitridation at 650 °C under ammonia gas flow for 3 h. As the control sample, Si@$TiO_{1-y}N_y$ was synthesized by the same procedure as mentioned above but was nitrided at 800 °C under NH_3 gas flow for 3 h. Meanwhile, to identify the phase transformation of TiO_2 according to the nitidation temperature, commercial TiO_2 powders (P25, Degussa) was used and was thermal treated for 5 h at various temperatures (600 to 950 °C) by NH_3 gas.

Characterization: The morphologies of the Si NPs, Si@TiO_2, and Si@TTN were characterized by FE-SEM (JSM-7001F, JEOL). The TEM bright-field images and EELS mappings were obtained by high-resolution transmission electron microscopy (HR-TEM, JEM-2200FS, JEOL). The XPS spectra were acquired using an X-ray photoelectron spectrometer (XPS, VersaProbe, Ulvac-PHI), and the crystal structures and phases of the products were characterized by XRD (D8 Advance, Bruker). The nitrogen adsorption and desorption isotherms of the samples were measured using the Brunauer–Emmett–Teller method (BELSORP-max, MicrotracBEL) after degassing at 150 °C for 12 h. For ex situ XRD measurements during first cycling, the coin cell was disassembled and then electrodes were washed with dimethyl carbonate. The electrodes were dried at 80 °C for 5 h and mounted on the XRD holder. To characterize the morphologies of the bare Si NPs, Si@TiO_2, and Si@TTN after 50 cycles, the electrodes were prepared in the same procedure as for ex situ XRD analysis and the active particles were transferred from the electrodes to an SEM/TEM holder.

Electrochemical Measurements: The Si@TTN composite was mixed with super-P and poly (acrylic acid) (Mw = 3 000 000, Sigma–Aldrich) at a weight ratio of 60 (active):20 (binder):20 (super-P) with distilled water. After mixing, the slurry was casted onto copper foil using the doctor blade method, and the electrode was dried in a vacuum oven at 80 °C overnight. The mass loading of the active material was 0.2–0.3 mg cm^{-2}. Electrodes of Si NPs and the Si@TiO_2 composite were also prepared by the same fabrication procedure. CR2032 coin cell assembly was performed in a drying room. A 1 M lithium hexafluorophosphate solution in a co-solvent of ethylene carbonate and diethyl carbonate (1:1 = v/v) containing 5 wt.% fluoroethylene carbonate (PANAX ETEC) was used as the electrolyte. The cells were assembled as half-cell type with lithium metal (Wellcos) as the counter electrode and separators (Tonen). Galvanostatic charge/discharge experiments were performed in the potential range of 0.01–1.5 V (vs Li/Li^+). CV was performed at a scan rate of 0.01 mV s^{-1}. For cycling performance testing, all samples were precycled once at 0.1 A g^{-1} within the voltage range of 0.01–1.5 V. EIS was performed in the frequency range of 0.01 to 100 kHz at an amplitude of 5 mV.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

anodes, coating, lithium-ion batteries, silicon nanoparticles

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